SafetyScan: A PPE Detection System for Workplace Safety, Week 2 Checkpoint Report

Progress Overview

By week 2, the project established the initial foundation for the SafetyScan PPE detection system. The work focused on data sourcing, splitting, and cleaning, and early YOLOv8 experiments to validate the object detection pipeline. The dataset was prepared in a YOLOv8-compatible format, and the confusion matrices from the early runs showed clearer predictions after the dataset was cleaned and reduced to the head and helmet classes.

1. Dataset Acquisition

The dataset used for this project is the Hard Hat Dataset sourced from Roboflow. This dataset is appropriate for the project because it contains annotated workplace images related to helmet usage and PPE compliance. The original dataset included annotated bounding boxes for the following classes: head, helmet, person.

Because the dataset was already exported in YOLO-compatible format, it included image files and corresponding label files that could be directly integrated into the YOLOv8 training pipeline.

3. Data Splitting

The dataset was split into training (train/) and validation (valid/) sets, with the validation set created out of about 10% of the training data. This separation lets the model learn from one portion of the data while being tested on another, helping track how well it generalizes.

Object detection models like YOLOv8 use the validation set during training to measure performance through metrics like precision, recall, and mAP.

4. Data Cleaning and Preprocessing

A major part of Week 2 focused on cleaning the dataset to reduce noise and improve model performance.

4.1. Removal of the “person” Class

The original dataset contained three classes: head, helmet, and person. Since the project focuses on helmet compliance, the person class was less relevant to the main objective.

To improve model focus, the person class was removed, leaving only:

- head
- helmet

This simplified the detection task and aligned the dataset more closely with the PPE detection objective.

4.2. Effect of Cleaning

The early confusion matrices suggest that removing the person class improved model focus. In train2, the person class was predicted less effectively than head and helmet. In train3, after removing the person class, the predictions became more focused on the two relevant classes. This indicates that cleaning the dataset helped minimize noise and made the training setup more appropriate for PPE detection.

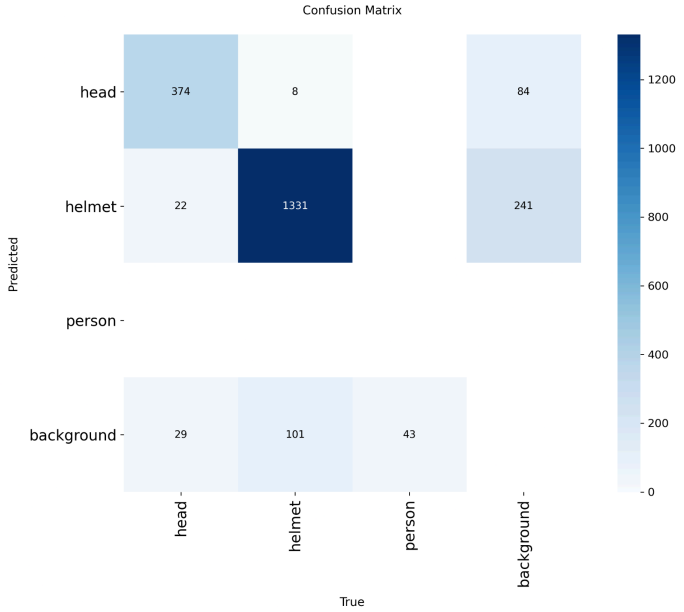
4.3. Dataset Structure

After cleaning, the dataset remained in YOLO format, with images stored in image folders and annotations stored in corresponding label folders using normalized bounding box coordinates. This structure is fully compatible with YOLOv8 for training, validation, and inference.

5. Initial Training Results

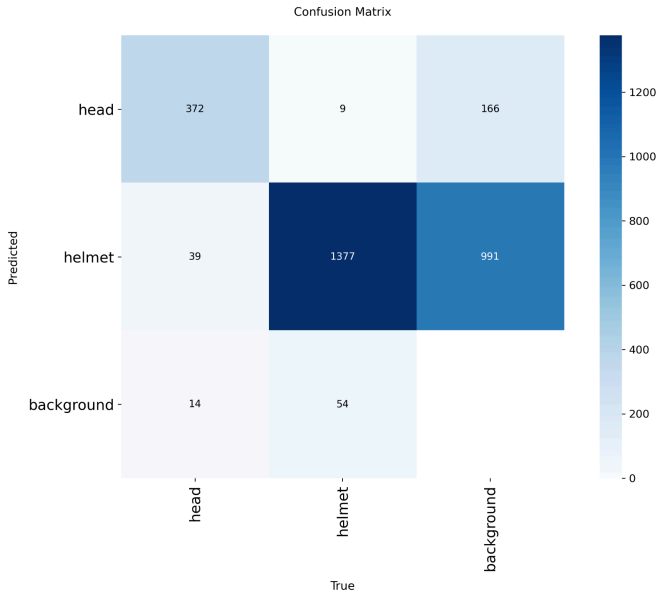
To verify that the training pipeline was working correctly, early YOLOv8 training runs were performed using the prepared dataset. The confusion matrices from train2 and train3 were used to examine how the model predicted the target classes before and after dataset cleaning.

Figure 1. Confusion Matrix for Train 2



The train2 confusion matrix shows that the model was able to detect the head and helmet classes, while the person class was predicted less effectively. Among the three classes, the helmet showed the strongest performance, and the person appeared to contribute weaker predictions. This suggests that the original three-class setup introduced additional noise into the detection task.

Figure 2. Confusion Matrix for Train 3



The train3 confusion matrix reflects the cleaned dataset after the person class was removed. In this run, the model focused only on head and helmet, resulting in a simpler class structure that is more appropriate for the PPE detection task. Overall, the comparison between train2 and train3 supports the decision to remove the person class. The confusion matrices suggest that the two-class setup is more suitable for the project because it reduces unnecessary class complexity and better matches the project's PPE detection objective.